ISYS1102/3479 – Database Applications

Week 1 – Introduction and Database Architecture

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RMIT UNIVERSITY

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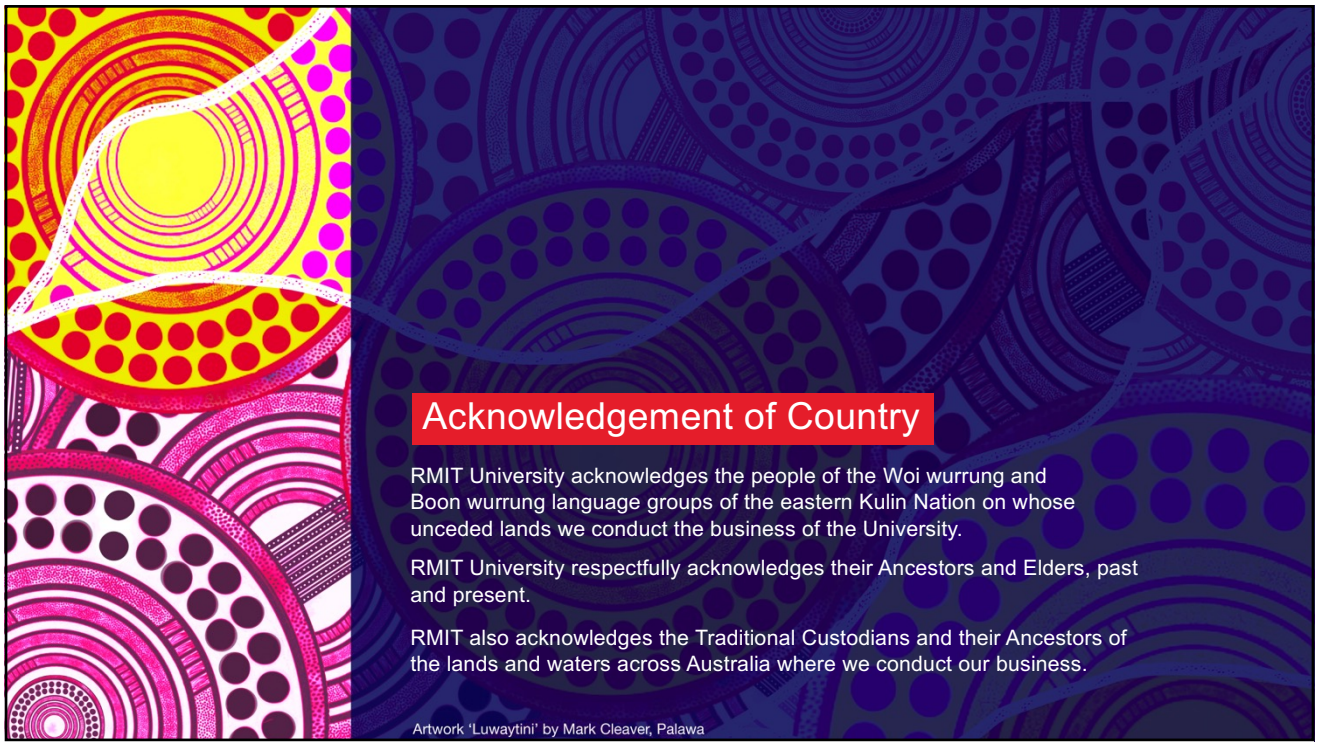

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What do you expect from this course?

Menti.com Code: 7563 9007

Link: https://www.menti.com/qnary4qvvd

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What is covered in this course?

https://miro.com/app/board/uXjVKwUF8FM=/

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Setting the Scene

A common complaint you heard about web sites:

- It takes too long to load my web page!
- More like than not, the culprit is the database back-end.
- Database performance matters – e.g. RMIT student database, search by name took approx. 13 seconds, where a search by student number would take less than a second.

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Setting the Scene

A common complaint you heard about web sites:

- 47% of consumers expect a web page to load in 2 seconds or less.
- Not only do consumers EXPECT your website to load in 2 seconds or less, but 40% of users will also abandon your website if it takes more than three seconds to load.
- So, DB Designers have an important role, making your database backend is efficient and queries are optimised.

(https://blog.kissmetrics.com/wp-content/uploads/2011/04/loading-time.pdf)

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How can we efficiently store data?

Storing large volumes of data in a methodical way is essential

- Traditional storage methods – flat files, spreadsheets – became insufficient and limited in many ways
- To meet the ever-increasing demand, since 1970s, databases and database management systems have been in use.

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Database Management Systems to the rescue!

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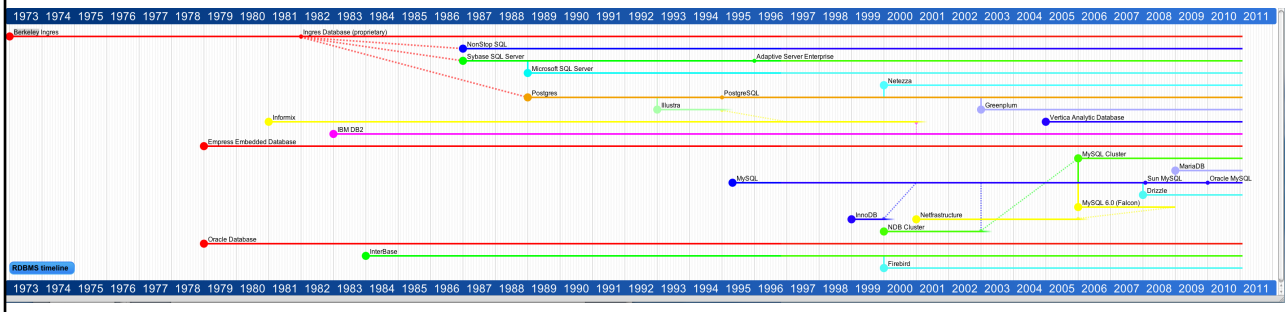
Database Management Systems to the rescue!

http://upload.wikimedia.org/wikipedia/commons/2/22/RDBMS_timeline.svg



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Database Management Systems to the rescue!

- Recently DBMS technology celebrated 50-year anniversary. Half a century is a long time in modern day technologies.
- So what is the secret to its longevity?
  - Its ability to evolve to keep pace with the ever-changing nature of data, applications, workloads, and deployment alternatives
  - Even if other database technologies, including object-oriented, graph, and document databases, have come into the market, relational databases continue to be the most popular DBMS.

Sources: https://www.oracle.com/database/50-years-relational-database

https://sqldbm.com/blog/from-the-birth-of-data-modeling-to-ai-milestones-and-timeline/

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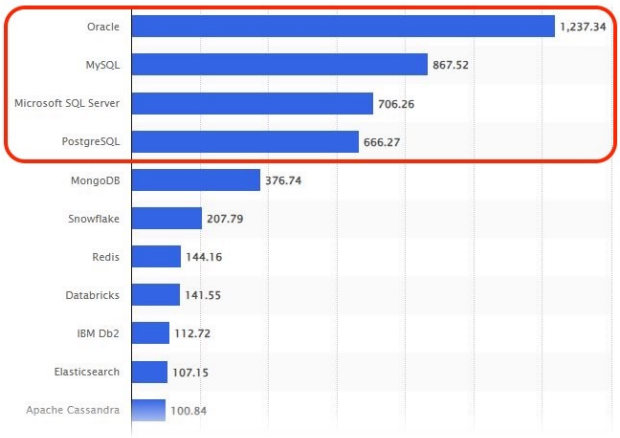
Which DBMS is most commonly used today?



https://www.statista.com/statistics/809750/worldwide-popularity-ranking-database-management-systems/

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What DBMS have you used?

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Link: https://www.menti.com/qnary4qvvd

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Comparing DBMS apples (and oranges)

https://en.wikipedia.org/wiki/Comparison_of_relational_database_management_systems

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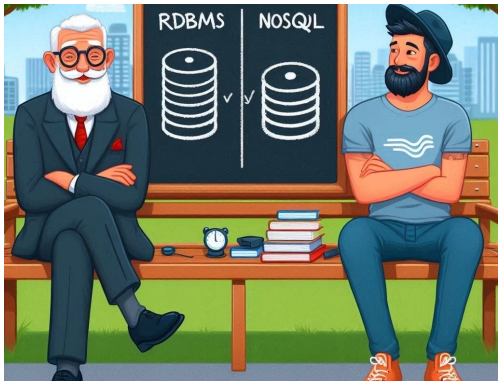
Non-relational (no-SQL) Databases

Re-invent the wheel?

- Data is everywhere. The digital transformation drives us to the frontiers that never reached.
- The digital transformation demands for rich content and big data!
- The relational database model, that catered the data management needs for almost half a century, has its inherent limitations.



Source: https://misapuntesde.com/2024/07/rdbms_vs_nosql.html

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Non-relational (no-SQL) Databases

Re-invent the wheel?

Relational Database Systems

- Oracle
- MS SQL Server
- MySQL
- PostgreSQL
- Many more…

Non-relational (no-SQL) Database Systems

- MongoDB
- Dynamo DB
- Regis
- Many more…

- Non-relational databases (AKA "not only SQL") store data differently than relational tables.
- They provide flexible schemas and scale easily with large amounts of big data and high user loads.
- four major types of NoSQL databases have emerged: document databases, key-value databases, wide-column stores, and graph databases.

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NoSQL – Not just one model

Document Databases	Graph Databases	Key-value Stores	Wide-column Stores
MongoDB	Azure Cosmos DB	Amazon DynamoDB	Azure Cosmos DB,
Google Cloud Firestore	InfiniteGraph	Azure Cosmos DB	Amazon DynamoDB,
Azure Cosmos DB	Apache Giraph	Couchbase	Cassandra
Apache CouchDB	Neo4J	Riak	Google Cloud Datastore
		Oracle NoSQL	HBase
			Bigtable

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SQL, NoSQL – Different Strengths

Data Model/ Functionality	Relational Database	Document Databases	Graph Databases	Key-value Stores	Wide-column Stores
Performance	variable	high	variable	high	high
Scalability	variable	high	variable	high	high
Flexibility	low	high	high	high	moderate
Data Integrity	high	low	low	low	low

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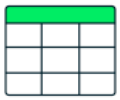
Comparing DB paradigms – relational vs non-relational



Relational Database

MongoDB

User table

ID	first_name	last_name	cell	city
1	Leslie	Yepp	8125552344	Pawnee

Hobbies table

ID	user_id	hobby
10	1	scrapbooking
11	1	eating waffles
12	1	working

```
{
   "_id": 1,
   "first_name": "Leslie",
   "last_name": "Yepp",
   "cell": "8125552344",
   "city": "Pawnee",
   "hobbies": ["scrapbooking", "eating
waffles", "working"]
}
```

- No need for joins
- No need for data normalization

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Database Performance. -- Focus in this course!!

Database often becomes the bottleneck in software performance. Refer to the original remarks I made at the beginning of this lecture.

Having an optimised database is essential for high performing systems.

https://danielfoo.medium.com/11-database-optimization-techniques-97fdbed1b627

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Database Performance

How?

Indexing

- Indexes are data structures that provide a quick lookup mechanism, significantly improving query performance. They work by creating a sorted data structure that allows the database engine to quickly locate the rows that satisfy a WHERE clause.
- While indexes speed up SELECT queries, they may slow down write operations, so it's crucial to strike a balance between read and write performance.
- Discussed next week

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Database Performance

How?

Good database design – Normalisation

- This process organises data to minimise redundancy and dependency, reducing the likelihood of data anomalies.
- By breaking down large tables into smaller, related ones, normalization ensures data consistency.
- However, it may lead to more complex queries.

- We have covered this in pre-req courses

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Database Performance

How?

Demormalisation

- While normalisation reduces redundancy, denormalisation introduces controlled redundancy to improve query performance, especially for read-heavy operations.
- This can involve adding redundant columns or tables strategically.
- MongoDB, a non-relational database system use this approach as its data model.
- We cover MongoDB in the second half of the semester.

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Database Performance

How?

Query Optimisation

- Optimise Queries: Regularly analyse and optimise frequently used queries.
- Use tools like EXPLAIN to understand the query execution plan and identify areas for improvement.
- Discussed in week 3.

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Database Performance

How?

Partitioning

- Partitioning involves dividing large tables into smaller, more manageable pieces.
- This can significantly improve query performance by allowing the database engine to work on smaller subsets of data, leading to faster query execution.
- Discussed in week 4

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Database Performance

How?

Concurrency Control

- Isolation Levels: Adjust isolation levels based on the requirements of your application.
- Isolation levels control the visibility of changes made by one transaction to other transactions.
- Choosing the appropriate isolation level is crucial for balancing consistency and performance.
- Discussed next week 7.

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House Keeping

Essential information for success

- Attendance is essential for successfully completing the course
- A laptop is required – either Windows or Mac is acceptable
  - Software:
    - Visual Studio Code
    - Mongo Compass
- Assignments -- 4 assessments in the course
  - Assignment 1 – relational database optimisation
  - Assignment 2 – Mongo DB design and implementation
  - Assignment 3 – Mid-semester test (MCQ and short-answer type)
  - Assignment 4 – Building a web database application using NodeJS and MongoDB.

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Planning for next week

Next Week (Monday is a public holiday).
Lectorial will be done offline – Recording will be posted
We will discuss Indexing and how indexing help us run queries faster.

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